Clinical & Operational Need Statement Formulation

Translating Real-World Observations into Solution-Neutral Biodesign Needs

Biodesign Innovation Framework

Indian Institute of Technology Kharagpur

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Biodesign Fundamentals: What is a Need?

Core Definition

A concise description of an unmet dilemma that identifies the **affected population**, the **core problem**, and the **desired outcome** *without embedding a solution*.

Key Characteristics:

- **Solution-Neutral:** Avoids premature technology selection (e.g., AI, apps, patches).
- **Measurable:** Focuses on impact and outcomes.
- **Actionable & Scoped:** Broad enough to invent, narrow enough to address the root dilemma.

The Biodesign Formula

A way to

[address problem / core dilemma]

in

[specific target population]

to / that

[achieve measurable desired outcome]

Pitfalls Avoided

× Solution embedded
× Built on a negative
× Too narrow/broad
× Symptom-only fix

Overview of Need Exploration Domains

No.	Project Domain	Core Clinical / Operational Focus
1	Maternal & Infant Nutrition	Personalized dietary guidance during pregnancy & lactation
2	Campus Zoonotic & Pest Hazards	Mitigation of stray animal injuries and venomous pest vectors
3	Campus Outpatient Care (BC Roy)	Acuity-based scheduling and academic-clinical workflow sync
4	Point-of-Need Medication Access	Therapeutic delivery for acutely incapacitated residential patients
5	Communal Dining & Metabolism	Menu preference alignment with macronutrient balance

Project 1: Maternal & Infant Nutrition – Need Exploration

Field Observations (PPTE)

- **People:** Pregnant women, new mothers, busy obstetricians, and pediatricians.
- **Process:** Constant recurring doubts on daily food safety, cultural restrictions, maternal chronic risks, and infant weaning diets.
- **Technology/Tools:** Ad-hoc phone calls or unverified internet searches.
- **Environment:** Home care between infrequent clinical appointments.

Problem Identification

Surface Problem: Pregnant women cannot reach doctors for every food question.

Deeper Problem: Absence of continuous, personalized, risk-adjusted dietary guidance causes nutritional anxiety, risk of consuming contraindicated foods, and outpatient clinician overload.

Project 1: Maternal & Infant Nutrition – Need Statement

Validated Biodesign Need Statement 1

"A way to provide personalized, clinically safe dietary and activity guidance in pregnant and postpartum women with varied baseline health profiles to ensure maternal-fetal nutritional safety and reduce routine clinical consultation load."

Problem ($\mathcal{P}$):
Lack of accessible, health-tailored dietary guidance and uncertainty over food/activity safety.

Population ($\mathcal{P}_{op}$):
Pregnant and lactating women with diverse maternal health/chronic conditions.

Outcome ($\mathcal{O}$):
Ensure maternal-fetal nutritional safety and minimize unneeded routine outpatient consultations.

Project 2: Stray Animals & Pest Reporting – Need Exploration

Field Observations (PPTE)

- **People:** Campus students/residents, aggressive or infected stray animals, venomous insects (wasps/bees) and snakes.
- **Process:** Sighting of animal injuries, rabies risks, or insect nests with no clear reporting channel.
- **Technology/Tools:** Word-of-mouth or unmonitored social forums.
- **Environment:** Expansive residential campus with dense foliage.

Problem Identification

Surface Problem: Students do not know who to call for stray dogs or wasps.

Deeper Problem: Lack of structured reporting and tracking mechanisms for zoonotic risks and venomous vectors leads to delayed administrative intervention, compromised campus hygiene, and physical injury risks.

Project 2: Stray Animals & Pest Reporting – Need Statement

Validated Biodesign Need Statement 2

"A way to streamline the identification, reporting, and tracking of aggressive/diseased animals and venomous pests in a residential campus community to accelerate institutional intervention and prevent public health hazards."

Problem ($\mathcal{P}$):
Uncoordinated reporting of animal-borne hygiene hazards, infections, and dangerous insect encounters.

Population ($\mathcal{P}_{op}$):
Residents and administrative municipal/veterinary departments of an academic campus.

Outcome ($\mathcal{O}$):
Accelerate response time for mitigation and reduce zoonotic/physical injury incidence.

Project 3: Advanced Clinical Scheduling – Need Exploration

Field Observations (PPTE)

- **People:** Outpatient student patients, healthcare providers, faculty advisors.
- **Process:** First-come-first-served physical queues without severity-based triage; manual excuse slip handoffs.
- **Technology/Tools:** Paper-based tokens or rudimentary sign-in sheets.
- **Environment:** Crowded primary healthcare center waiting halls during peak lecture hours.

Problem Identification

Surface Problem: Waiting times at the campus health center are too long.

Deeper Problem: Un-stratified patient intake leads to delayed care for acute illnesses, inefficient clinician utilization, and academic-attendance penalties for students due to disconnected administrative logging.

Project 3: Advanced Clinical Scheduling – Need Statement

Validated Biodesign Need Statement 3

"A way to triage and coordinate outpatient consultations in university students seeking campus healthcare to reduce clinical waiting times, prioritize acute medical needs, and prevent academic schedule disruption."

Problem ($\mathcal{P}$):
Inefficient consultation queues lacking clinical acuity triage and academic schedule synchronization.

Population ($\mathcal{P}_{op}$):
University students and institutional primary care providers.

Outcome ($\mathcal{O}$):
Minimize patient waiting time, prioritize urgent cases, and streamline institutional excuse logging.

Project 4: Point-of-Need Medication Access – Need Exploration

Field Observations (PPTE)

- **People:** Acutely ill students (severe gastroenteritis, high fever, acute pain), hall caretakers, pharmacists.
- **Process:** Physically walking to a central dispensary while non-ambulatory or experiencing active symptoms.
- **Technology/Tools:** In-person prescription counter pickup.
- **Environment:** Multi-story residential halls distanced from the campus hospital.

Problem Identification

Surface Problem: Sick students cannot go to the pharmacy to buy medicine.

Deeper Problem: Physical barrier to accessing prescribed therapeutics during acute, immobilizing non-emergency conditions results in delayed therapy, physical distress, and disease transmission risks.

Project 4: Point-of-Need Medication Access – Need Statement

Validated Biodesign Need Statement 4

"A way to facilitate timely point-of-need access to essential medications and rehydration for acutely incapacitated residential students to accelerate symptomatic relief and prevent physical deterioration."

Problem ($\mathcal{P}$):
Inability of acutely symptomatic, non-ambulatory individuals to physically retrieve prescribed medication.

Population ($\mathcal{P}_{op}$):
Acutely ill students residing in campus residential hostels.

Outcome ($\mathcal{O}$):
Accelerate symptom management, avoid dangerous delays, and eliminate the strain of physical transit.

Project 5: Communal Dining Nutrition – Need Exploration

Field Observations (PPTE)

- **People:** Hostel residents, catering committees, mess cooks.
- **Process:** Static menu preparation leading to high food wastage, students skipping meals or consuming low-nutrition junk food.
- **Technology/Tools:** Rigid quarterly paper menus.
- **Environment:** Institutional mass kitchens and dining halls.

Problem Identification

Surface Problem: Students do not like the mess food and want to vote for dishes.

Deeper Problem: Misalignment between institutional meal planning and diner preferences leads to skipped meals, dietary deficiencies (protein/micronutrients), and massive food expenditure wastage.

Project 5: Communal Dining Nutrition – Need Statement

Validated Biodesign Need Statement 5

"A way to align institutional meal planning with resident dietary preferences in communal hostel dining to increase regular meal participation and maintain balanced macronutrient intake."

Problem ($\mathcal{P}$):
Disconnection between catering provision and consumer palatability causing meal skipping.

Population ($\mathcal{P}_{op}$):
Institutional student residents relying on communal hostel dining facilities.

Outcome ($\mathcal{O}$):
Maximize dining engagement and ensure adequate daily protein and carbohydrate nutrition.

Methodological Validation of Formulated Needs

Project	Common Pitfall Avoided	Outcome Metric	Innovation Zone
1. Maternal Nutrition	No embedded app/AI; not limited to single disease	Maternal-fetal safety; reduced routine triage visits	Mixed Need (Clinical & behavioral)
2. Pest & Stray Hazards	No embedded phone hotline/app; focuses on hazard mitigation	Response latency; reduction in bite/sting incidents	Incremental / Operational workflow
3. Clinical Scheduling	Avoids "booking software"; addresses clinical prioritization	Wait time (minutes); academic conflict rate (%)	Incremental / Operational triage
4. Medicine Access	Avoids "drone/courier delivery"; focuses on point-of-need	Time to first dose; reduced dehydration/relapse	Mixed / Logistics & health delivery
5. Mess Management	Avoids "voting app"; focuses on nutrition & satisfaction	Meal participation rate; protein/carb adherence	Incremental / Behavioral nutrition

Key Takeaways: Why Solution-Neutrality Matters

The Power of Need Scoping

- **Broadens Solution Space:** By stating *"a way to facilitate timely point-of-need access"* instead of *"a delivery app"*, we allow solutions ranging from hall-level first-aid lockers to buddy logistics or automated dispensaries.
- **Focuses on End Outcomes:** Ensures teams measure success by **clinical safety, wait times, and health impact**, not software features.

Next Steps in Biodesign Cycle

1. **Phase 2: Need Screening** (Stakeholder analysis, market size, clinical fundamentals).
2. **Phase 3: Concept Generation** (Brainstorming multiple solutions per need).
3. **Phase 4: Concept Selection** (Filtering concepts against regulatory and feasibility constraints).